Risk and Adaptivity in Certifying Machine-Learning Model Comparisons

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Abstract

Background. When verification records are incomplete, an auditor needs to know whether further readings will make a model comparison precise enough. We ask when choosing the next reading after seeing an earlier answer improves that guarantee under a fixed budget.

Methods. We bound model-loss differences over all compatible records. We classify scalar width penalties and prove a shared-query separation with matching lower bounds. We then check precision certificates on frozen forecast comparisons and replay the policies on stored answers.

Results. A shared-query construction has optimal adaptive mean joint width $1/2$, versus $2/3$ for randomized batching, with matching lower bounds. For scalar width, the two minimax risks agree universally exactly for nondecreasing concave penalties. On 165 of 819 original blocks and 57 of 279 external blocks, two shared adaptive readings guarantee joint half-width precision, whereas optimal randomized-batch worst-case success is exactly 50%. These are precision-success guarantees, distinct from mean-width risk. On the certified subsets, recorded success is 100% versus 65.45% and 64.91% for the selected optimal batch mixtures. Complete-panel replay gives 7.81- and 11.11-percentage-point gains over fixed worst-width batches, and 2.20- and 2.15-point gains over greedy adaptation. Under illustrative 1% internal errors with exact endpoints, 45 external blocks retain an adaptive guarantee that every randomized batch fails to match. Capacity information reduces this advantage, and measured public-API retrieval favors bulk.

Conclusions. These guarantees depend on the supplied bounds and access rules. The shared-budget evidence is retrospective; physical meter accuracy and improved downstream model choice are not established.

Keywords: artificial intelligence; machine learning; adaptive information; minimax risk; optimal recovery; robust optimization.

1 Introduction

A model audit may need additional measurements before it can justify a comparison. The auditor can retrieve the full record, choose a batch of readings in advance, or choose later readings after

seeing earlier answers. We call the last option adaptive querying. A sharp interval is the smallest interval containing the loss difference for every record compatible with the available information. We ask when feedback improves the guarantee that this interval will be sufficiently narrow. Width limits how much the reported loss difference can vary across compatible records. The application requires retrieval to halve that range: a declared uncertainty budget for reporting a comparison, not a validated proxy for choosing a winner.

We study affine scalar contrasts on nonempty compact convex uncertainty sets, with queries selected from a fixed finite dictionary of linear measurements. This formulation includes paired squared-loss comparisons, overlapping linear constraints, valid coordinate bounds, signed targets, and fixed query costs. Batch and adaptive policies have the same hard budget. The unknown record is fixed before the algorithm's random seed is drawn; probabilities describe acquisition randomization, not a distribution assumed for the missing truth. Fixed bounded measurement errors can be included by enlarging the hidden record when the resulting uncertainty set is compact and convex.

The practical decision is which retrieval protocol can meet the auditor's required precision. We use fixed model predictions and choose which verification readings to retrieve. A certificate checks whether the resulting comparison meets the requirement for every compatible record. This differs from improving the predictor itself.

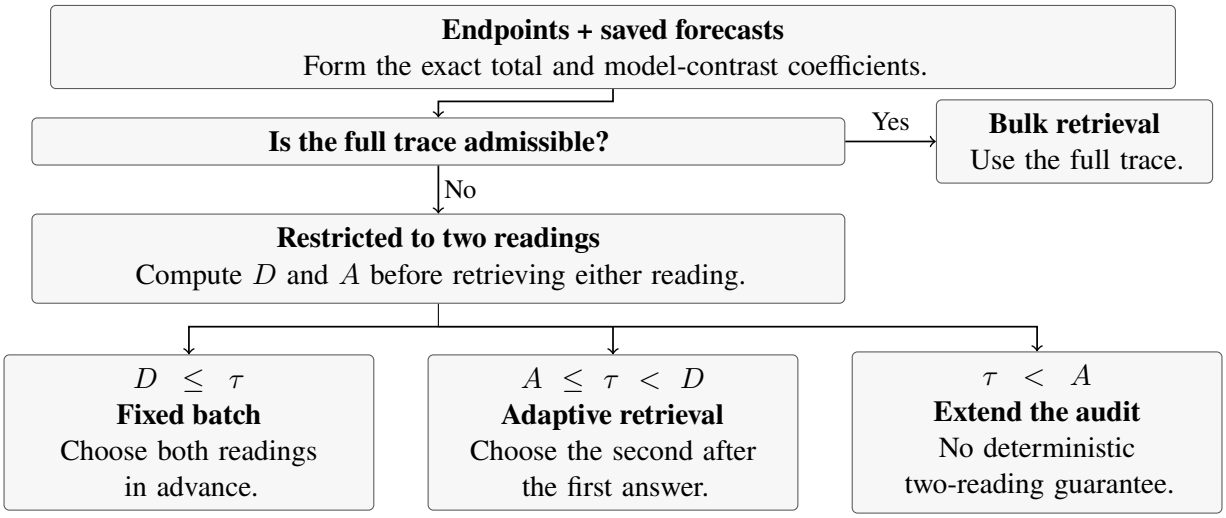


Figure 1: Choosing an audit protocol. For the exact-total, noiseless prefix model, let $J_Q = \max_r W_{r,Q}/W_{r,0}$ over contrasts with positive initial width. Let $D = \min_Q \sup_y J_Q(y)$ and $A = \min_{P \text{ det. adaptive}} \sup_y J_{Q_P(y)}(y)$ under the same two-reading budget. We fix the target τ before retrieval; the main application uses $\tau = 1/2$. Since $A \leq D$, the three restricted-access branches exhaust deterministic guarantee possibilities. Randomized precision success is assessed separately. Bulk is preferred on the measured public API when its full response is admissible (Section 3.4).

Our primary contribution is the mathematical classification and the explicit shared-query separation in Table 1. The classification uses established minimax reasoning and Jensen's inequality; we supply its coordinate-query converse. A separate construction proves an exact adaptive advantage for the largest normalized width across contrasts. The supporting dimension-dependent precision result adapts sparse-policy bounds (Kurtz, 2026).

The data answer a related, more specific question: can the same two readings make all six forecast comparisons precise? We identify 165 original and 57 external blocks with a guaranteed adaptive success probability of one and an optimal randomized-batch success probability of one half. These exact instance certificates are the principal application evidence. The complete 819- and 279-block replays test selected policies on the stored answers; their empirical success rates do not estimate the constructed mean-risk gap. This is a retrospective extension using frozen data, not an additional independent sample. Capacity and error controls test the information model; the measured API workload identifies bulk retrieval as preferable when its full response is admissible.

Table 1: Claims, objectives and evidence. A guarantee over compatible records differs from success on the single stored record in each audit.

Claim	Objective	Evidence and scope
Scalar classification	Worst expected penalty of one interval's width	Theorem 1: universal equality exactly for nondecreasing concave penalties.
Shared-query separation	Worst expected largest normalized width	Theorem 4: constructed optima $1/2$ versus $2/3$, each with matching bounds.
Certified meter instances	Worst-case probability that every width meets its target	165/819 and 57/279 blocks: adaptive optimum one versus batch optimum $1/2$; worked proof in Section 2.6.2.
Complete-panel replay	Precision success on stored answers	Section 3.1: fixed selected policies on all 1,098 blocks; incomplete batch-precision bounds remain explicit.
Retrieval interface	Recorded requests and I/O time	Section 3.4: bulk wins on a three-task public-API workload; this does not measure deployment savings.

The paper develops the classification and sharp examples, then computable two-query guarantees and their cumulative-meter specialization. The evaluation reports fixed protocols, recorded replay, uncertainty sensitivity and actual retrieval measurements. Online Resource 1 provides extended proofs, the separate compact coordinate-query solver, historical controls, complete case ledgers and executable verification.

1.1 Related work and the contribution boundary

Partial identification studies conclusions that survive incomplete observations (Manski, 2005; Maia Polo et al., 2024). For model comparison, cancellation in paired squared losses leaves an affine function of the unknown truth; this is established loss structure rather than a new loss function (Patton, 2011). Active evaluation and model selection acquire labels for estimating performance or choosing among predictors (Sawade et al., 2012; Hara et al., 2024; Kossen et al., 2021; Okanovic et al., 2025). Our target is the entire sharp interval of a fixed-record contrast after exact observations. No estimator of the population risk, surrogate label distribution, or new forecasting architecture is introduced.

The certificate width is the established local diameter of information (Novak and Woźniakowski, 2008, Section 4.1). Osipenko (2019, Theorems 2–3 and 5) also studies when adaptation cannot

improve noisy scalar recovery. Those criteria concern deterministic reconstruction error and the existence of an optimal nonadaptive linear reconstruction; the present criterion instead classifies randomized penalties of the complete terminal information-fiber width. Classical deterministic results already establish strong limits on adaptivity: on convex symmetric classes with linear information, Novak and Woźniakowski (2008, Theorem 4.4) construct nonadaptive information with no greater global diameter and at most twice the reconstruction radius. For scalar targets the radius comparison is exact (Novak and Woźniakowski, 2008, Theorem 4.5). A modern formulation of the factor-two bound appears in Krieg et al. (2025, Proposition 4.1).
Nonsymmetric convex uncertainty itself is also established: Novak (1995) compares adaptive and nonadaptive optimal-recovery errors using relations between Gelfand and Bernstein widths. The classical objective is reconstruction error, whereas our randomized penalty applies to the full terminal compatible interval. Our classification allows nonsymmetric convex sets and concerns randomized penalties of the terminal local diameter, including a converse and exact precision and moment separations. Its mean-risk interchange applies Sion (1958, Theorem 3.4); the concave-penalty upper bound uses Jensen's inequality.
For fixed deterministic scalar information, the optimal worst-case reconstruction error is half the largest fiber width. The distinction here concerns randomized local-width penalties and the order of averaging and worst-case evaluation. Specifically, Krieg et al. (2025, Section 2) and Krieg and Ullrich (2026, Section 9.1, Eq. (9.15), author manuscript v2) study $\inf_A \sup_{y\in F} \mathbb{E}_\omega \|T(y) - A_\omega(y)\|$, the error of a selected reconstruction. Here the random transcript instead incurs the full compatible-interval width $W_{Q_\omega(y)}(y)$. Writing its scalar local radius as $r_{Q_\omega(y)}(y) = W_{Q_\omega(y)}(y)/2$, our mean risk is $2\sup_y \mathbb{E}_\omega r_{Q_\omega(y)}(y)$. For a randomized batch, this generally differs from $2\mathbb{E}_\omega \sup_y r_{Q_\omega}(y)$, twice the expected global radius discussed in Krieg and Ullrich (2026, Eq. (2.4), Remark 6.1 and Section 6.1, author manuscript v2). The midpoint reconstruction has pointwise error at most the local radius, but its error at the actual record need not equal that radius. Consequently, neither replacing actual reconstruction error by interval width nor interchanging the expectation and supremum preserves the optimization problem. The present theorem concerns the terminal local-width risk. The fixed measurement dictionary is substantive: if every linear functional were directly queryable at the same cost, a scalar affine target could be determined by querying its linear part once.
Information-dependent decisions are also established in adjustable robust optimization (Wei and Zhang, 2024; Vayanos et al., 2026). The dimension-dependent precision bound below adapts the sufficient-policy result of Kurtz (2026, Theorem 2 and Example 5); its query interpretation and sharp examples are made explicit. Weighted-oracle randomized optimization and dynamic-program LP constructions are established (Kawase and Sumita, 2024; Martin et al., 1990). The supporting aggregate solver contributes a structured hard-budget batch representation and dual-completion certificate (Online Resource 1, S5/S7). Secondary optimization among tied robust solutions follows an established concern (Iancu and Trichakis, 2014).
Adaptive sampling of cumulative information has close precedents. Gaillard et al. (2023, Section 4.1.1, Corollary 1 and Theorem 3) study certified scalar integral estimation from sequential evaluations of a monotone function. For monotone contrast coefficients, summation by parts relates the register problem to weighted monotone integration. Our instance test pairs an adaptive precision guarantee with an adverse distribution limiting every randomized batch under the same information interface.
Kuboyama (2026, Theorem 1 and Proposition 1) gives exact model-choice certificates after

adaptive binary labels and compares shared-label paired bounds with separate intervals. That work also distinguishes hindsight certificate size from acquisition requirements (Theorem 2 and Proposition 3). Our application uses constrained continuous targets, while the central classification concerns the penalty of terminal sharp width. These precise distinctions, rather than certification or shared comparisons generally, define the contribution relative to model-choice and information-discovery frameworks. The shared-query objective is the established weighted ℓ_∞ local diameter of a vector target, not a new uncertainty measure. Our additional result is its exact two-prefix separation and computable guarantees for joint forecast comparisons. Neither shared observations nor adaptive vector recovery alone is claimed as new.

2 Methods

2.1 Model Comparisons and Sharp Information

Let $a, b \in \mathbb{R}^N$ be two fixed prediction vectors for the same scheduled targets. Their paired mean-squared-error difference is

$$D(y) = \frac{1}{N}\sum_{i=1}^{N}\{(a_i - y_i)^2 - (b_i - y_i)^2\} = d + c^\top y, \qquad c_i = \frac{2(b_i - a_i)}{N}. \tag{1}$$

All scheduled targets remain in the denominator. Initially observed values, documented aggregate measurements and other defensible constraints define the feasible set F. Any permitted archive record that directly reconstructs a target is incorporated before this set is declared. Sharpness is always relative to that declared information set.

The attainable interval of D on F is an identification interval for this record, not a sampling confidence interval. A negative upper endpoint certifies that a has smaller squared loss for every completion; a positive lower endpoint certifies b. An interval containing zero cannot certify a strict winner. Subsequent exact measurements restrict F to a fiber and therefore cannot widen its sharp interval. A prescribed width threshold instead asks for a precision guarantee, regardless of the interval's location.

The mathematical examples below also describe paired squared-loss widths. Given any contrast coefficient c, nonnegative predictions $a_i = N\max(-c_i, 0)/2$ and $b_i = N\max(c_i, 0)/2$ realize that coefficient in (1). The induced constant changes the interval's position, but not its width. Thus the constructions have a model-comparison interpretation without being measurements of prediction performance.

2.2 Information, terminal width, and risk

Let $F \subset \mathbb{R}^n$ be nonempty, compact, and convex, and let $T(y) = c^\top y + d$ be the scalar quantity to be audited. A fixed finite collection of exact linear measurements is available. Write L_Q for the measurement matrix of a selected set Q, and let $\mathcal{Q}$ be the nonempty finite family of admissible final sets, available to both policy classes. Admissibility is determined before observing answers; for example, $\sum_{j\in Q} k_j \le B$ for nonnegative fixed costs k_j. Coordinate observations correspond to rows of the identity matrix. The results do not require nonnegative targets or disjoint aggregates.

For a feasible record y, define the information fiber and its sharp target width by

$$F_Q(y) = \{z \in F : L_Q z = L_Q y\}, \qquad W_Q(y) = \max_{z \in F_Q(y)} T(z) - \min_{z \in F_Q(y)} T(z). \tag{2}$$

The extrema exist by compactness, and $0 \le W_Q(y) \le D_T$, where $D_T = \max_F T - \min_F T$. This is the local diameter of the scalar information fiber. The optimized quantity is the full fiber's diameter, rather than the actual error of a selected point estimate. An additive constant in T does not affect W_Q. In particular, a paired squared-loss contrast is affine in the unknown verification targets and fits this formulation.

A randomized batch policy draws $Q \sim \pi$ before seeing answers. An adaptive policy selects successive measurements and a stopping decision from the answers already observed and an independent internal random seed. Every execution must terminate after finitely many queries at a set in $\mathcal{Q}$. Query and stopping rules are jointly measurable in the observed history and seed; observations are exact. For a finite-valued, nondecreasing function $\phi : [0, \infty) \to \mathbb{R}$, define

$$R^{\phi}_{\text{batch}} = \inf_{\pi \in \Delta(\mathcal{Q})} \sup_{y \in F} \mathbb{E}_{Q \sim \pi} \phi(W_Q(y)), \qquad R^{\phi}_{\text{adaptive}} = \inf_{\mathcal{P}} \sup_{y \in F} \mathbb{E} \phi(W_{Q_{\mathcal{P}}(y)}(y)). \tag{3}$$

The hidden record is fixed before the random seed is drawn. Thus the supremum is outside the expectation. When measurement error is represented by additional hidden coordinates, one fixed error vector is part of that record for the entire execution. An adversary choosing a new error after each query is a different information model. These objectives concern terminal width under a hard budget, rather than expected requests until a winner is certified. The expectations are well defined: the widths are Borel measurable by Proposition 1, and a finite-valued monotone ϕ is Borel measurable and bounded on $[0, D_T]$.

For a required precision $\tau \ge 0$, let $\mathcal{C}$ be either the randomized batch or randomized adaptive policy class, and write $Q_{\mathcal{P}}(y, \omega)$ for the terminal set under internal seed ω. Define its optimal worst-case precision-success probability by

$$\begin{aligned} S^{\mathcal{C}}_{\tau} &= \sup_{\mathcal{P} \in \mathcal{C}} \inf_{y \in F} \Pr_{\omega} \big\{ W_{Q_{\mathcal{P}}(y,\omega)}(y) \le \tau \big\} \\ &= 1 - R^{\phi_\tau}_{\mathcal{C}}, \qquad \phi_\tau(w) = \mathbf{1}\{w > \tau\}, \end{aligned} \tag{4}$$

where $R^{\phi_\tau}_{\mathcal{C}}$ denotes the corresponding risk in (3). The identity follows by complementing the success event and reversing each supremum or infimum under subtraction from one; it does not require an optimal policy to attain either bound. The probability concerns the algorithm's seed at a fixed hidden record, and equality at the threshold counts as success. Reaching this precision does not require the interval to exclude zero, so it need not certify a winning model.

Proposition 1 (Concavity and the transcript fiber)**.** *For every fixed Q, W_Q is nonnegative, finite, concave, and upper semicontinuous on F. For every adaptive execution, its terminal information fiber is precisely $F_Q(y)$ for its terminal queried set Q.*

Proof. Choose maximizing and minimizing endpoints $u_k, v_k \in F_Q(y_k)$, $k = 1, 2$. For $0 \le \lambda \le 1$, convexity of F and linearity of the measurements make $\lambda u_1 + (1-\lambda)u_2$ and $\lambda v_1 + (1-\lambda)v_2$ feasible endpoints at $\lambda y_1 + (1-\lambda)y_2$. Their target difference is $\lambda W_Q(y_1) + (1-\lambda)W_Q(y_2)$, proving concavity. For upper semicontinuity, take $y_k \to y$ and a subsequence attaining the limsup

of the widths. Compactness supplies a further subsequence whose endpoint pairs converge to $u, v \in F_Q(y)$. Its limiting difference is at most $W_Q(y)$.

For the transcript claim, fix the internal random seed. If no query is made, both fibers are F. Otherwise, any $z \in F_Q(y)$ gives the same answer to the first queried measurement. Inductively, equal previous answers produce the same next measurement and the same next answer. The stopping decision is also unchanged. Thus every completion agreeing with all queried answers follows the same entire transcript. Conversely, any completion consistent with that transcript must agree on every queried answer. Query choices and the stopping time therefore impose no additional restrictions on the fiber beyond those exact observations. This argument is pointwise in the seed; no conditioning on a positive-probability seed event is required. □

2.3 Which width penalties can benefit from adaptation?

The next result uses classical minimax theory. The contribution is the resulting classification for sharp information-fiber widths and its coordinate-query converse, rather than a new minimax or duality theorem. The proof has two parts. For concave penalties, a single hardest record limits every final query set, including those chosen adaptively; Jensen's inequality supplies a batch policy meeting that limit. For any nonconcave penalty, a three-coordinate construction realizes a strict midpoint violation as an adaptive advantage. These arguments establish the universal claim; the later numerical checks test implementations on particular inputs.

Theorem 1 (Universal classification of width penalties)**.** *Let $\phi : [0, \infty) \to \mathbb{R}$ be finite-valued and nondecreasing. Randomized batch and randomized adaptive querying have equal minimax ϕ-risk for every problem satisfying Section 2.2 if and only if ϕ is concave. More precisely, put*

$$\rho = \min_{\pi \in \Delta(\mathcal{Q})} \max_{y \in F} \sum_{Q \in \mathcal{Q}} \pi_Q W_Q(y). \tag{5}$$

If ϕ is concave, then

$$R^{\phi}_{\text{adaptive}} = R^{\phi}_{\text{batch}} = \phi(\rho). \tag{6}$$

Every batch policy attaining (5) *simultaneously attains this equality for every such concave ϕ. If ϕ is not concave, a strict adaptive improvement already occurs on a three-coordinate compact convex polytope, with all coordinates queryable at unit cost and a hard budget of two.*

Proof. The function $\sum_Q \pi_Q W_Q(y)$ is affine and continuous in π, and concave and upper semicontinuous in y. Sion's minimax theorem (Sion, 1958) therefore yields

$$\rho = \max_{y \in F} \min_{Q \in \mathcal{Q}} W_Q(y). \tag{7}$$

All values are finite. Writing $G(\pi) = \max_{y \in F} \sum_Q \pi_Q W_Q(y)$, the bound $|G(\pi) - G(\pi')| \le D_T \|\pi - \pi'\|_1$ proves continuity. Thus the minimum in (5) is attained on the compact simplex. The maximum in (7) is attained because the minimum of finitely many upper semicontinuous functions is upper semicontinuous on compact F. Hence there is one record $y^\star \in F$ such that $W_Q(y^\star) \ge \rho$ for every admissible Q. By Proposition 1, every execution of every adaptive policy has terminal width at least ρ at this same record. Monotonicity gives the lower bound $\phi(\rho)$.

Let $\pi^\star$ attain (5). Concavity, Jensen's inequality, and monotonicity give, for every $y \in F$,

$$\sum_Q \pi_Q^\star \phi(W_Q(y)) \le \phi\left(\sum_Q \pi_Q^\star W_Q(y)\right) \le \phi(\rho).$$

This proves the upper bound and simultaneous optimality. No continuity of ϕ at zero is needed; the finite-mixture form of Jensen's inequality suffices. For example, $\phi(w) = \mathbf{1}\{w > 0\}$ is concave on $[0, \infty)$ despite its jump at zero.

For necessity, a nonconcave, nondecreasing finite-valued ϕ must have a strict midpoint-concavity violation: there exist $0 \le l < h$ such that, with $m = (l + h)/2$,

$$\phi(m) < \{\phi(l) + \phi(h)\}/2. \tag{8}$$

Indeed, if midpoint concavity held, for $0 \le x \le z$ and dyadic $q_k \uparrow \theta$ its dyadic inequalities and monotonicity would give

$$\phi((1-\theta)x + \theta z) \ge \phi((1-q_k)x + q_k z) \ge (1-q_k)\phi(x) + q_k \phi(z).$$

Taking the limit only on the right proves full concavity without a continuity assumption, a contradiction. Set $d_0 = h - l$ and consider

$$F_{l,h} = \{(a, b, c) : 0 \le a \le 1,\ 0 \le b \le l + d_0 a,\ 0 \le c \le h - d_0 a\}, \qquad T = ha + b + c. \tag{9}$$

All three coordinates are available at unit cost, with budget two. The widths after querying $\{a, b\}$ and $\{a, c\}$ are respectively $h - d_0 a$ and $l + d_0 a$. An adaptive policy first observes a and then queries the coordinate with the larger remaining capacity. Its terminal width is at most m, so its risk is at most $\phi(m)$. At $(a, b, c) = (1/2, 0, 0)$, these two widths equal m, while the width after querying $\{b, c\}$ is h. Thus every admissible terminal set has width at least m, and the adaptive optimum is exactly $\phi(m)$.

Now put equal adversarial mass on $(0, 0, 0)$ and $(1, 0, 0)$. Each batch $\{a, b\}$ or $\{a, c\}$ has average penalty $\{\phi(l) + \phi(h)\}/2$. The batch $\{b, c\}$ has width h at both records, so its average penalty is no smaller. A batch using fewer coordinates cannot improve upon an extension to two coordinates, because more observations only shrink the fiber. Every randomized batch policy consequently has worst-case risk at least $\{\phi(l) + \phi(h)\}/2 > \phi(m)$. This is the required strict separation. □

The universal quantifier is essential. A nonconcave penalty permits an adaptive advantage somewhere in this class; it need not yield an advantage for every feasible set. The theorem covers heterogeneous fixed costs and overlapping constraints at the level of information theory. It does not imply a polynomial-time optimizer for an arbitrary feasible set or budget.

2.3.1 A three-coordinate example

Take $0 \le a \le 1$, $0 \le b \le a$, $0 \le c \le 1 - a$, and $T = a + b + c$, with two unit-cost queries. The widths for batches $\{a, b\}$, $\{a, c\}$, and $\{b, c\}$ are $1 - a$, a, and $1 - b - c$. Querying a first and then c if $a \le 1/2$, or b otherwise, leaves width $\min(a, 1 - a) \le 1/2$.

An equal mixture of $\{a, b\}$ and $\{a, c\}$ attains the batch values. The endpoint records $(0, 0, 0)$ and $(1, 0, 0)$ give the corresponding batch lower bounds. At $(1/2, 0, 0)$ every admissible terminal set has width at least $1/2$, establishing the adaptive moment lower bounds. The precision separation holds for every threshold $1/2 \le \tau < 1$.

Table 2: Exact constructed-example values. These are mathematical guarantees, not forecasting improvements measured on public data.

Criterion	Best randomized batch	Best adaptive policy
Worst expected width	$1/2$	$1/2$
Worst expected squared width	$1/2$	$1/4$
Worst-case success, width $\leq 3/4$	$1/2$	1

2.4 Growing moment and precision separations

For $K \geq 2$, let $a_1, \ldots, a_K \geq 0$ sum to one and impose $0 \leq b_j \leq a_j$. The actual queryable coordinates are $a_1, \ldots, a_{K-1}, b_1, \ldots, b_K$; a_K is determined by the first $K-1$ capacities. Define

$$T = \sum_{i<K} a_i + \sum_{j=1}^{K} b_j, \qquad B = 2K - 2. \tag{10}$$

This is a compact convex polytope in $2K-1$ dimensions, and every coordinate has the same unit cost.

Theorem 2 (Sharp growing separations)**.** *For* (10)*, the optimal worst expected width is* $1/K$ *for both randomized batch and adaptive policies. For every* $p > 1$*,*

$$R^{w^p}_{\text{batch}} = 1/K, \qquad R^{w^p}_{\text{adaptive}} = 1/K^p. \tag{11}$$

For required width $\tau = 1/K$*, the best worst-case success probabilities are* $1/K$ *for randomized batches and* 1 *for adaptive policies.*

Proof. It suffices to consider batches using the full budget: adding observations cannot increase width. Such a batch omits exactly one coordinate. Omitting b_j leaves width a_j; omitting a_i, $i < K$, leaves width $a_i + a_K - b_i - b_K$. The latter follows because the remaining feasible range of a_i is $[b_i, a_i + a_K - b_K]$ and its coefficient in T is one. An adaptive policy queries all explicit capacities and then all b_j except one having the smallest capacity. Its width is $\min_j a_j \leq 1/K$. At the record with all $a_j = 1/K$ and all $b_j = 0$, every full-budget batch has width at least $1/K$. This proves the adaptive lower bounds and the mean-width lower bound.

The randomized batch policy omitting each b_j with probability $1/K$ has expected width exactly $1/K$. Its expected pth power is $K^{-1}\sum_j a_j^p \leq 1/K$, since $0 \leq a_j \leq 1$ and $p > 1$. For the reverse moment bound, average over the K records $a = e_j$, $b = 0$. A batch omitting a b_j has average pth-power width $1/K$; one omitting an explicit a_i has average $2/K$. Therefore every batch distribution has worst-case moment at least $1/K$.

For precision, omitting a uniformly random b_j succeeds with probability at least $1/K$ at every record, because at least one capacity is no greater than $1/K$. To prove the matching upper bound, use K records indexed by j, with $b = 0$,

$$a_j = \frac{1}{2K}, \qquad a_i = \frac{2K-1}{2K(K-1)} \quad (i \neq j).$$

These capacities sum to one. Only the batch omitting b_j reaches width at most $1/K$ at record j: every other b-omitting batch has larger width, and every a-omitting batch has width $a_i + a_K > 1/K$. If q_j is the probability of omitting b_j, worst-case success is at most $\min_j q_j \le 1/K$. The adaptive policy always succeeds. The precision upper bound uses these interior capacity records; checking only vertices would not establish it. □

The moment ratio is K^{p-1}, while the ratio of worst-case success probabilities at precision $1/K$ is K. These separations use a structured constructed feasible set and a budget querying all but one coordinate. They establish the possible difference between objectives; they do not establish comparable gains at small relative budgets in a deployment or in the historical forecasting study.

2.5 Computing policies and hardest records on polytopes

Let $F = \{y \in \mathbb{R}^n : Ay \le b\}$ be nonempty and compact, and let $T(y) = c^\top y + d$. Equalities can be represented by paired inequalities. The constant d does not affect width. The first formulation allows any declared finite family $\mathcal{Q}$ of legal batches; L_Q denotes the matrix of their exact linear measurements.

2.5.1 An enumerative hardest-record formulation

Introduce a common record y, endpoint witnesses u_Q, v_Q for each $Q \in \mathcal{Q}$, and a scalar η. Consider

$$\begin{aligned} \max \quad & \eta \\ \text{subject to} \quad & Ay \le b, \quad Au_Q \le b, \quad Av_Q \le b \quad (Q \in \mathcal{Q}), \\ & L_Q u_Q = L_Q y, \quad L_Q v_Q = L_Q y \quad (Q \in \mathcal{Q}), \\ & \eta \le c^\top (u_Q - v_Q) \quad (Q \in \mathcal{Q}). \end{aligned} \tag{12}$$

For fixed y, the independent witnesses for batch Q can attain its sharp width $W_Q(y)$. Thus the largest feasible η at that record is $\min_Q W_Q(y)$, and the optimum of (12) is ρ. Its common record is a certificate that every legal batch leaves width at least ρ there.

The nonnegative dual multipliers π_Q of the last constraints form an optimal randomized batch policy. Indeed, the free variable η forces $\sum_Q \pi_Q = 1$. Dualizing these constraints first yields

$$\min_{\pi \in \Delta(\mathcal{Q})} \max_{y, u_Q, v_Q} \sum_Q \pi_Q c^\top (u_Q - v_Q) = \min_{\pi \in \Delta(\mathcal{Q})} \max_{y \in F} \sum_Q \pi_Q W_Q(y) = \rho,$$

where the endpoint feasibility constraints remain in the inner maximization. Ordinary linear-programming strong duality supplies the equality and the policy. In the implementation, a separate LP fixes the extracted π and maximizes its expected width, checking the upper bound independently of the constraints involving η.

With $M = |\mathcal{Q}|$ batches, the formulation has $n(1 + 2M) + 1$ variables. It is polynomial in the explicitly enumerated batch family, which may be exponential in the number of available measurements. It is a small-instance benchmark and certificate generator; the aggregate-specific compact formulation avoids this enumeration in its structured setting.

2.5.2 An exact deterministic two-query algorithm

Assume every one of the $n \geq 2$ coordinates is queryable at unit cost and the hard budget is two. An unused query can be added without increasing width, so considering two distinct coordinates includes policies that stop earlier. For a first query i and feasible answer t, define

$$g_{ij}(t) = \max\{c^\top(u - v) : Au \leq b,\ Av \leq b,\ u_i = v_i = t,\ u_j = v_j\}, \qquad j \neq i. \tag{13}$$

This maximizes the terminal width over all feasible records having first answer t, after second query j. The best deterministic policy beginning with i therefore has worst width $\rho_i = \max_t \min_{j \neq i} g_{ij}(t)$.

Proposition 2. *For each i, ρ_i is the value of the following single linear program, with one endpoint pair (u^j, v^j) for each $j \neq i$:*

$$\begin{aligned} \max \quad & \eta \\ \textit{subject to} \quad & Au^j \leq b, \quad Av^j \leq b && (j \neq i), \\ & (u^j)_i = (v^j)_i = t, \quad (u^j)_j = (v^j)_j && (j \neq i), \\ & \eta \leq c^\top(u^j - v^j) && (j \neq i). \end{aligned} \tag{14}$$

The optimal deterministic adaptive two-query worst-width value is $\min_i \rho_i$.

Proof. At fixed feasible t, each endpoint pair has only its own constraints and can attain $g_{ij}(t)$. All inequalities on η are feasible precisely up to their smallest attainable upper bound, $\min_{j \neq i} g_{ij}(t)$. Maximizing over the shared t proves the first claim. For any deterministic policy beginning with i, its chosen second coordinate at answer t has conditional worst width at least $\min_j g_{ij}(t)$. Conversely, choosing a minimizing j after observing t attains that value. Finite tie breaking gives a measurable selection because these polyhedral value functions are piecewise linear on their feasible interval. Finally, minimize over the first coordinate. □

For rational input, n LPs of polynomial description size choose the first coordinate; after its answer, $n - 1$ conditional LPs implement the second choice. This is a polynomial-time deterministic two-query procedure in the standard bit-complexity model, including the encoding length of rational query answers. For arbitrary exact answers, the policy is still defined by the conditional LPs; the polynomial bit-complexity statement applies to rationally encoded inputs and answers. It does not solve arbitrary-budget randomized precision-risk optimization or minimize expected requests until certification. No grid of possible first answers is used.

2.5.3 A precision consequence of sparse-policy bounds

The dimension bound below adapts established sparse-policy reasoning. Kurtz (2026, Theorem 2 and Example 5) gives the corresponding dimension-plus-one policy bound for continuous concave scenario costs, including conditional information discovery. Conditional terminal widths fit that framework on polytopes. The direct Helly proof below also allows compact convex F, for which these widths are upper semicontinuous but need not be continuous, and translates the sparse cover into precision success under uniform randomization.

Theorem 3 (A dimension-dependent batch guarantee, adapted from sparse-policy bounds)**.** *Let F be nonempty, compact and convex, and let $T(x) = c^\top x + d$. Fix a first-stage exact linear observation $L_0 x$, whose feasible-answer set $I = L_0 F$ has affine dimension r. After this answer, a deterministic policy chooses one of finitely many second-stage linear measurement batches $L_1, \dots, L_m$. Every combined batch (L_0, L_j) must obey the same declared hard budget, and the eligible second-stage batches must not depend on the first answer. Suppose the policy's terminal sharp interval has width at most τ for every $x \in F$. Then some randomized nonadaptive policy supported on at most $r+1$ of these combined batches has worst-case precision success at least $1/(r+1)$. In particular, a sure-precision policy using two scalar queries yields a randomized-batch success guarantee of at least $1/2$, even when its conditional worst widths are not affine functions of the first answer.*

Proof. For $t \in I$, define the conditional worst terminal width

$$g_j(t) = \max\{c^\top(u - v) : u, v \in F,\ L_0 u = L_0 v = t,\ L_j u = L_j v\}.$$

The feasible set is nonempty and compact, so the maximum is attained. The expression is exactly the largest terminal sharp width among records with first answer t: an endpoint pair with equal first and second answers belongs to a common terminal fiber, and the endpoints of every such fiber are feasible in this maximization.

Each g_j is concave on I. Indeed, convex combinations of optimizing endpoint pairs at t_1, t_2 are feasible at the corresponding convex combination of answers, and attain the same convex combination of their objective values. The function is also upper semicontinuous: for a convergent sequence of answers, compactness gives convergent subsequences of optimizing endpoints, whose limits are feasible at the limiting answer. Consequently, g_j is finite and measurable.

For a fixed first answer, the policy chooses a single second-stage batch; its sure guarantee therefore implies $\min_j g_j(t) \le \tau$ throughout I. The strict failure sets

$$H_j = \{t \in I : g_j(t) > \tau\}$$

are convex by concavity, and $\bigcap_{j=1}^m H_j = \varnothing$. Finite Helly's theorem (Amenta et al., 2016, Theorem 1.1), applied in the affine hull of I, gives a subfamily J of at most $r+1$ sets with empty intersection. This finite form of Helly's theorem does not require the sets to be closed; thus using strict failure sets preserves success at equality with τ.

Choose uniformly among the combined batches indexed by J. For every fixed record x, its answer $t = L_0 x$ lies outside at least one H_j, so that batch's actual terminal width is at most $g_j(t) \le \tau$. Its precision success probability is therefore at least $1/|J|$, and hence at least $1/(r+1)$, for every record. □

The empty second-stage action includes stopping after the first stage. Fixed unequal costs are allowed if all combined batches obey the budget; the action dictionary and latent record are fixed before randomization. The K-capacity construction in Theorem 2 is sharp with first-answer dimension $r = K - 1$ and optimal batch success $1/K$. For scalar first answers, at most two second-stage actions suffice to retain the optimal sure-width guarantee. On polytopes, test each singleton or pair of actions with the shared-first-answer LP in Proposition 2; there are at most $\binom{m}{2} + m$ candidates. This sparse policy need not minimize conditional width at every individual answer. Helly-based sparsification for convex-function optimization is classical (Rockafellar,

1965, Section 2, Theorem 1). The contribution here is its certificate-risk interpretation and sharpness through admissible information queries, rather than a new general sparse-policy theorem.

2.5.4 Adaptive precision from cumulative-counter queries

Consider a six-hour block with nonnegative hourly increments $z = (z_1, \dots, z_6)$ and retained cumulative registers C_0, C_6, all expressed in the same energy units. Writing $E = C_6 - C_0$, the initial feasible set is $F_E = \{z \geq 0 : \sum_{h=1}^{6} z_h = E\}$. For two frozen forecast vectors a, b, their block MSE contrast is

$$D(z) = d + c^\top z, \qquad d = \frac{1}{6}\sum_{h=1}^{6}(a_h^2 - b_h^2), \quad c_h = \frac{2(b_h - a_h)}{6}.$$

The permitted query at boundary $k \in \{1, \dots, 5\}$ returns C_k, equivalently the prefix sum $s_k = C_k - C_0$. The budget is at most two distinct internal registers per pair and block. This interface retrieves cumulative registers. The coordinate-increment recovery controls are documented separately in Online Resource 1, S2–S4. The register queries are exact linear measurements covered by the general information model. Adaptive scalar certification also appears in monotone integral estimation (Gaillard et al., 2023, Section 4.1); here the target is an affine contrast of interval increments under a fixed hard budget.

For integer boundaries $0 \leq p < q \leq 6$, define $r(p, q) = \max_{p<h\leq q} c_h - \min_{p<h\leq q} c_h$. With queried boundaries Q, let $0 = q_0 < \cdots < q_m = 6$ include the outer boundaries and Q, and put $s_0 = 0$, $s_6 = E$. Nonnegativity and the segment totals give the sharp width

$$W_Q(z) = \sum_{\ell=1}^{m}(s_{q_\ell} - s_{q_{\ell-1}})r(q_{\ell-1}, q_\ell). \tag{15}$$

Both endpoints are attainable by placing each segment's mass at a minimum- or maximum-coefficient hour. The initial width is $W_0 = Er(0, 6)$. Tasks with $W_0 = 0$ are already precise and require no queries.

Proposition 3 (Two best-side splits for an arbitrary block)**.** *Let $n \geq 3$, $E > 0$, $x \geq 0$, $\sum_{h=1}^{n} x_h = E$, and let the target be $d + \sum_h c_h x_h$. Each query returns one internal prefix sum. Write $r(p, q) = \max_{p<h\leq q} c_h - \min_{p<h\leq q} c_h$. For first boundary k, put $\ell_k = r(0, k)$ and $r_k = r(k, n)$. Define each of the following quantities whenever its corresponding side has an eligible second boundary:*

$$\alpha_k = \min_{1\leq j<k} \max\{r(0, j), r(j, k)\}, \qquad \beta_k = \min_{k<j<n} \max\{r(k, j), r(j, n)\}.$$

Choose a minimizing boundary on each side, breaking ties by smallest index. When both sides are nonempty, for normalized first answer $t = s_k/E$, the optimal conditional worst width is

$$E \min\{\alpha_k t + r_k(1 - t),\ \ell_k t + \beta_k(1 - t)\}.$$

Thus only one best boundary on each side is needed, independently of t. If $\Delta_k = \ell_k + r_k - \alpha_k - \beta_k > 0$, the switch is

$$t_k^\star = \frac{r_k - \beta_k}{\Delta_k} \in [0, 1],$$

and the worst width for this first boundary is

$$A_k = E \max\left\{\alpha_k, \beta_k, \frac{\ell_k r_k - \alpha_k \beta_k}{\Delta_k}\right\}.$$

Use the best right split below the switch and the best left split above it; either is valid at equality. If $\Delta_k = 0$, the two lines coincide and $A_k = E\max\{\ell_k, r_k\}$. At $k = 1$ only the right split exists and $A_1 = E\beta_1$; at $k = n-1$ only the left split exists and $A_{n-1} = E\alpha_{n-1}$. The optimal deterministic two-query guarantee is

$$A = \min_{1 \le k < n} A_k. \tag{16}$$

Proof. All second queries left of k have the common right-side contribution $r_k(1-t)$; minimizing their remaining term t times a split range selects the same minimizing boundary for every $t \ge 0$. The right-side argument is identical. Moreover $\alpha_k \le \ell_k$ and $\beta_k \le r_k$. Hence $\Delta_k \ge 0$, and when it is positive the displayed intersection belongs to $[0, 1]$. The maximum of the lower envelope occurs at an endpoint or that intersection. Its endpoint values are β_k, α_k, and its intersection value is the displayed quotient. If $\Delta_k = 0$, both nonnegative differences vanish, so the lines coincide. At an outer eligible boundary the single-hour side has zero coefficient range, giving the stated one-line values. □

All contiguous ranges can be built in $O(n^2)$ arithmetic operations and stored in $O(n^2)$ space. Scanning the candidates for every α_k, β_k costs another $O(n^2)$ operations; the closed forms then cost $O(n)$. This is an arithmetic-operation bound for the exact-total, uncapped two-prefix problem, not a bit-complexity or noisy-polytope claim. For $n = 2$, the single internal register determines both increments; for $n = 1$ the total already determines the only increment.

An upper certificate for randomized-batch precision. Returning to the six-hour case, there are ten two-boundary batches. Let $z^{(h)} = Ee_h$ be the six feasible single-hour mass profiles, and define $M_{hQ}(\tau) = \mathbf{1}\{W_Q(z^{(h)}) \le \tau\}$. The finite-profile program

$$U(\tau) = \min_{w \ge 0,\, \sum_h w_h = 1,\, u} u \quad \text{subject to} \quad \sum_h w_h M_{hQ}(\tau) \le u \quad \text{for every batch } Q \tag{17}$$

upper-bounds the worst-case precision success of *every* randomized batch policy. Indeed, its minimum success over F_E is at most its success averaged over these profiles under w, which is at most $\max_Q \sum_h w_h M_{hQ}$. This remains valid for batches with fewer queries: adding an unused register cannot increase width. The exported w therefore provides a checkable upper certificate. The finite profile set need not contain a worst completion for an indicator objective, so $U(\tau)$ is not asserted to equal the full precision optimum in general. Below, write $S_B^\star = S_\tau^{\text{batch}}$ and $S_A = S_\tau^{\text{adaptive}}$ for the precision-success values defined in (4), at this fixed instance and threshold. If $A \le \tau$ and $U(\tau) < 1$, the adaptive policy succeeds for every completion, whereas no randomized batch can do so. In floating-point classification, batch widths within $10^{-9}W_0$ above the threshold count as success, and adaptive certification requires a margin of $10^{-9}W_0$; exact constructed boundary cases are treated analytically.

An exact counter example. Take $E = 1$ and $c = (0, 0, 1, 1, 0, 1)$, so $W_0 = 1$. Query boundary 4 first, obtaining $t = s_4$. Batch $\{2, 4\}$ has width $1 - t$, while batch $\{4, 5\}$ has width t. Querying 5 when $t < 1/2$ and 2 otherwise thus guarantees width $\min\{t, 1 - t\} \le 1/2$. Conversely, at $z = e_5$ only batch $\{4, 5\}$ attains width below 1; that batch has width 1 at $z = e_2$. No batch succeeds at both profiles for any threshold $\tau < 1$. Weights $w_2 = w_5 = 1/2$ in (17) give $U \le 1/2$. An equal mixture of $\{2, 4\}$ and $\{4, 5\}$ succeeds with probability at least $1/2$ on every completion when $\tau \ge 1/2$. Consequently, for every $\tau \in [1/2, 1)$ the optimal adaptive worst-case success is 1 and the optimal randomized-batch worst-case success is $1/2$.

Corollary 1 (The two-register half-success floor)**.** *If a deterministic two-register policy guarantees width at most τ, a randomized batch supported on at most two batches has worst-case precision success at least $1/2$. Thus an adverse-profile upper bound $U(\tau) = 1/2$ proves the exact batch optimum for such an instance.*

Proof. Apply Theorem 3 with scalar first answer. For the exact-total prefix model the conditional functions are affine: their good sets are prefixes or suffixes of $[0, 1]$. The longest good prefix and earliest good suffix cover all answers; mix their batches equally, breaking geometric ties by smallest second-query index. □

Corollary 2 (Deciding whether feedback is necessary)**.** *In the cumulative-counter model, let $\mathcal{B}$ contain the ten two-register batches, and define*

$$D = \min_{Q \in \mathcal{B}} \max_{z \in F_E} W_Q(z) = \min_{Q \in \mathcal{B}} \max_{1 \le h \le 6} W_Q(Ee_h).$$

Let A be the exact deterministic adaptive value in (16)*, and let $U(\tau)$ be the finite-profile upper certificate in* (17)*. For every $\tau \ge 0$,*

$$D \le \tau \quad \Longleftrightarrow \quad U(\tau) = 1 \quad \Longleftrightarrow \quad S_B^\star = 1.$$

Consequently, the available forecasts and outer registers determine three mutually exclusive decisions before any optional retrieval:

1. *If $D \le \tau$, a fixed batch guarantees the requested precision.*
2. *If $A \le \tau < D$, feedback enables a deterministic guarantee that every randomized batch fails to match.*
3. *If $\tau < A$, no deterministic adaptive policy with two register queries guarantees the requested precision.*

In the second case, $1/2 \le S_B^\star \le U(\tau) \le 5/6$. The lower bound uses Corollary 1; when $U(\tau) = 1/2$, the randomized-batch optimum is exactly $1/2$.

Proof. For a fixed batch, (15) is a linear function of z. It therefore attains its maximum over the energy simplex at a concentration profile Ee_h, which proves the expression for D. If $D \le \tau$, one batch succeeds at all six profiles and, by linearity, at every completion. Thus $S_B^\star = 1$ and $U(\tau) = 1$. If $D > \tau$, every batch fails at at least one profile. The uniform distribution on the six profiles consequently assigns every batch success probability at most $5/6$. It is a feasible adverse distribution in (17), so $S_B^\star \le U(\tau) \le 5/6 < 1$. This proves both equivalences. Adaptation includes fixed batches, hence $A \le D$. The three cases then follow from the defining optimality of A and D. □

This decision rule uses only the forecast coefficients, retained total, query dictionary and requested precision. It identifies when a sequential query response is necessary for a guarantee, without measuring retrieval latency, financial cost or the frequency of adverse energy profiles. The finite-profile program is decisive for whether batch success equals one; it remains only an upper bound on intermediate success values unless a matching guarantee is supplied.

2.6 Exact shared-query separations: policies, lower bounds, and verifiable certificates

An auditor comparing several models can reuse each reading across their loss contrasts. Let $W_{r,Q}(y)$ be the sharp width for contrast r after the common transcript Q, and let $W_{r,0} > 0$ be its initial width. Define

$$J_Q(y) = \max_r W_{r,Q}(y)/W_{r,0}.$$

Then $J_Q(y) \le 1/2$ means that every requested comparison has reached half its original width. Already determined contrasts can be omitted. This is the local diameter of the vector contrast map in a weighted ℓ_∞ norm, within the established vector-recovery framework (Krieg et al., 2025); shared-label model certification also has close prior work (Kuboyama, 2026).

Theorem 4 (Feedback reduces mean joint width with two shared queries)**.** *Let $F = \{x \in \mathbb{R}^6_+ : \sum_h x_h = 1\}$, and allow at most two queries from $P_k x = \sum_{h \le k} x_h$, $k = 1, \dots, 5$. For the two contrasts with coefficient vectors*

$$c^{(1)} = (1/2, 0, 1, 1/2, 1/2, 0), \qquad c^{(2)} = (2/3, 2/3, 1, 1/3, 0, 1),$$

both initial widths are one. Let $J_Q(x)$ be the larger terminal sharp width and let $R^J_{\mathcal{C}} = \inf_{P \in \mathcal{C}} \sup_{x \in F} \mathbb{E} J_{Q_P(x)}(x)$. Then

$$R^J_{\text{randomized adaptive}} = \tfrac{1}{2} < \tfrac{2}{3} = R^J_{\text{randomized batch}}.$$

Proof. A batch partitions the six coordinates into segments. Its scalar width is the sum, over segments, of segment mass times coefficient range. After querying boundary 3 with answer t, the worst remaining joint widths for second queries 5 and 2 are, respectively,

$$G_{35}(t) = \max(t, 1/3), \qquad G_{32}(t) = \max(1/2, 1 - t).$$

Query 5 when $t \le 1/2$ and query 2 otherwise. This guarantees $J \le 1/2$. At the single feasible record $x^* = (0, 0, 1/2, 0, 0, 1/2)$, all ten final two-query sets have $J_Q(x^*) \ge 1/2$; their values are the first row below. Here and below ij means boundaries $\{i, j\}$.

Q	12	13	14	15	23	24	25	34	35	45
$J_Q(x^*)$	1	3/4	5/6	1/2	1/2	5/6	1/2	3/4	1/2	1/2
$\mathbb{E}_\mu J_Q$	2/3	2/3	2/3	2/3	2/3	13/18	5/6	5/6	2/3	2/3

For a fixed internal random seed, every completion agreeing with the queried answers follows the same adaptive transcript and stopping rule. Its terminal interval is therefore the final-set interval used above. Fewer queries cannot decrease width. Thus every randomized adaptive policy has expected width at least $1/2$ at x^*.

For the batch upper bound, mix $12, 23, 35$ with probabilities $1/6, 1/3, 1/2$. Its expected joint widths at $e_1, \ldots, e_6$ are $(2/3, 2/3, 2/3, 2/3, 2/3, 1/2)$. Each J_Q is a maximum of linear segment widths and hence convex; these vertex values bound the expectation throughout F by $2/3$. For the reverse bound, place adverse masses $\mu = (1/3, 0, 1/6, 1/6, 1/3, 0)$ on those vertices. The table's second row shows that every batch has expectation at least $2/3$, and therefore so does every batch mixture. Its worst-record expectation is no smaller. □

This is a constructed separation for *mean joint width*, not a precision-success probability. It does not follow from the scalar classification: a maximum of concave local widths need not be concave. The argument uses established vector recovery and finite-game bounds (Krieg et al., 2025); its claim is the explicit shared-prefix separation. The two contrasts can be realized with a common forecast baseline $a = 0$ and $b^{(r)} = 3c^{(r)}$ in a six-target paired MSE. Adding other contrasts would define a different joint objective. For the uncapped exact-total simplex, each batch's scalar widths are linear in the energy allocation. Thus J_Q is convex. Its worst mixture expectation is attained at a concentration vertex, making the finite-game LP exact for randomized-batch mean risk. The deterministic two-query planner instead minimizes the worst joint width after feedback. Its conditional objective is the maximum of the contrast-specific affine widths; checking all line intersections gives the exact optimum. The two objectives remain distinct. A deterministic adaptive value below the randomized-batch lower bound establishes an adaptive advantage; the reverse inequality does not establish optimality among adaptive policies. The vertex argument does not transfer to precision indicators, caps or measurement error. S8 provides the formulations and verification details.

2.6.1 Planning two shared prefix queries

We plan one pair of readings for all requested contrasts. The inputs are m normalized coefficient rows $c_r \in \mathbb{R}^n$, each with range one, an exact total $E > 0$, and all internal prefix queries $P_k x = \sum_{h \le k} x_h$, where $x = y/E \ge 0$ and $\sum_h x_h = 1$. We assume $n \ge 3$ and a budget of two distinct queries. Define the range over a consecutive segment by $\Delta_r(a{:}b) = \max_{a \le h \le b} c_{rh} - \min_{a \le h \le b} c_{rh}$.

Suppose the first query is k and its answer is t. For a proposed second query $j \ne k$, the conditional worst width of contrast r is $\alpha_{kjr} t + \beta_{kjr}(1 - t)$, where

$$(\alpha_{kjr}, \beta_{kjr}) = \begin{cases} (\max\{\Delta_r(1{:}j), \Delta_r(j+1{:}k)\}, \Delta_r(k+1{:}n)), & j < k, \\ (\Delta_r(1{:}k), \max\{\Delta_r(k+1{:}j), \Delta_r(j+1{:}n)\}), & j > k. \end{cases}$$

Within each side of the first boundary, a worst-case allocation places its mass in the segment with the larger coefficient range. Maximizing over records and over the finite contrast family commutes, so

$$g_{kj}(t) = \max_r \{\alpha_{kjr} t + \beta_{kjr}(1 - t)\}, \qquad A = \min_k \max_{t \in [0,1]} \min_{j \ne k} g_{kj}(t)$$

is the optimal deterministic two-query worst-width guarantee.

Algorithm: two shared queries, exact-total simplex

1. For each first boundary k, form every affine line $\ell_{kjr}(t) = (\alpha_{kjr} - \beta_{kjr})t + \beta_{kjr}$.
2. Form B_k from 0, 1, and every intersection in $[0, 1]$ of two nonparallel lines, including lines belonging to different second queries.
3. Compute $A_k = \max_{t \in B_k} \min_{j \neq k} \max_r \ell_{kjr}(t)$. Choose the smallest boundary index attaining $\min_k A_k$.
4. Retrieve that boundary's counter value and calculate its normalized prefix mass t using the retained endpoint total.
5. Choose the smallest $j \neq k$ attaining $\min_j g_{kj}(t)$, retrieve it, and calculate all terminal segment-width certificates using the two shared answers.

Between consecutive points of B_k, the ordering of all underlying lines is fixed. The maximum within each group and then the minimum over groups therefore select one affine line on that interval. Its maximum is at an endpoint. This proves that the finite calculation covers every possible first answer; it is not a grid approximation. Equal or coincident lines are handled by the stated index rule.

The mathematical algorithm uses exact comparisons. The reported six-interval implementation uses the same construction in floating-point arithmetic: it treats values within 10^{-12} of a minimum as tied and chooses the smallest index, both offline and online. Its intersection routine skips slope differences of magnitude at most 10^{-14}; a 257-point grid is an additional numerical check. Exact rational branch and adversarial certificates support the reported exact assurance claims. The algorithm above addresses deterministic worst width on the uncapped, exact-total simplex. Bounded-error and capacity analyses use the separate conditional calculations described in their sensitivity protocols; randomized mean-width and precision-success comparisons retain their own optimization objectives.

Why the scalar precision bound does not apply to joint widths. Theorem 3 assumes one scalar target and an adaptive policy that guarantees precision. Its conditional worst widths are concave in the first answer, so their strict failure sets are convex. Here g_{kj} is a maximum of affine contrast widths and is convex; its failure set may be disconnected. The scalar Helly argument therefore does not transfer to joint precision. The joint advantage requires the matching certificates below. Figure 2(a) illustrates the conditional curves and the selected second query on one recorded task.

2.6.2 An inspectable certificate from the external panel

Consider HPM 1 on 5 July 2025, 12:00–18:00: the first matched-half case in the frozen external ledger. Normalize each of its six forecast contrasts to coefficient range one and write $x = y/E$ for the hourly allocation divided by its known block total. Thus $x \geq 0$, $\sum_h x_h = 1$, and joint precision means $J \leq 1/2$. The contrast order is daily–boosting, daily–MLP, daily–ridge, boosting–MLP, ridge–boosting, and ridge–MLP. The normalized coefficient rows have the following rational lower enclosures; every entry other than zero or one lies within 10^{-4} above its

displayed value, and zero and one are exact:

$$C \in L + [0, 10^{-4}) \quad \text{entrywise}, \qquad L = \begin{pmatrix} 1 & .2928 & .0571 & .1475 & 0 & .0026 \\ 1 & .3251 & .6303 & .4169 & .1256 & 0 \\ 1 & .1466 & .0899 & .0242 & 0 & 0 \\ 0 & .3884 & 1 & .6778 & .6220 & .5040 \\ 1 & .4847 & .0140 & .3095 & 0 & .0061 \\ .2093 & .3546 & 1 & .7180 & .2280 & 0 \end{pmatrix}.$$

With $Q = 12$ and $R = 24$, their normalized width-coefficient vectors have the form

$$v_{Qr} = (0, 0, a_r, a_r, a_r, a_r), \qquad v_{Rr} = (b_r, b_r, d_r, d_r, e_r, e_r).$$

The following rational upper enclosures are obtained from the exact saved coefficients; each actual range lies between the displayed value minus 10^{-4} and that value, with zero and one exact.

Contrast r	$a_r \le$	$b_r \le$	$d_r \le$	$e_r \le$
1	0.1476	0.7072	0.0905	0.0027
2	0.6304	0.6749	0.2134	0.1257
3	0.0900	0.8534	0.0657	0
4	0.4960	0.3885	0.3222	0.1180
5	0.3096	0.5153	0.2955	0.0062
6	1	0.1453	0.2820	0.2281

Batch lower guarantee. For each contrast pair (r, s) choose

$$\lambda_{rs} = \begin{cases} 1, & r \in \{1, 3, 4, 5\}, \\ 1/2, & r = 2, \\ (1/3, 1/3, 3/7, 0, 1/4, 0)_s, & r = 6. \end{cases}$$

The table directly verifies all 36 conditions

$$\max\{(1 - \lambda_{rs})b_s, \lambda_{rs}a_r + (1 - \lambda_{rs})d_s, \lambda_{rs}a_r + (1 - \lambda_{rs})e_s\} \le 1/2.$$

Thus Q and R cannot both fail: a failing contrast from each would make the indicated convex combination exceed $1/2$. Their equal mixture therefore succeeds with probability at least $1/2$ for every feasible record.

Matching upper bound. Place half the adverse mass at e_2 and half at $(e_4 + e_5)/2$. Batch 12 has joint widths zero and one there. Every other two-query batch leaves coordinate 2 in a segment containing coordinate 1 or 3. The saved coefficients satisfy $|c_1^{(3)} - c_2^{(3)}| \ge 0.8533$ and $|c_2^{(6)} - c_3^{(6)}| \ge 0.6453$, so all other batches fail at e_2. Every batch therefore succeeds on at most half of this adverse mass; every randomized batch has worst-case success at most $1/2$. The full-continuum batch optimum is exactly $1/2$.

Adaptive certificate and verification. The checked adaptive policy queries boundary 2, then 4 below its switch $\theta \in [0.5226, 0.5227]$ and 1 above it. The table bounds the two enlarged branch domains by 0.47743079 and 0.4774, respectively. These conservative bounds exceed the exact envelope peak, about 0.47734, plotted in Figure 2(a). Hence its success is guaranteed, even allowing either action inside the displayed switch enclosure. From the extracted package root, run the following commands; they require no optimization solver:

```
cd work/risk_adaptivity_revision
python joint_certification/proof_check_joint.py
python joint_certification/precision_batch_check.py
python joint_certification/precision_adaptive_link.py
python joint_certification/check_main_inspectable_certificate.py
```

The first three verify the constructed theorem and all 57 matching half-success certificates within the complete 279-block external panel. The last verifies this example's source hashes, 36 coefficient enclosures, 24 range enclosures, 36 multipliers, 20 adverse widths, and both branch domains. Full coefficients and the rational switch remain in its output. The original panel's 165 certificates require the documented raw-data regeneration route; these default commands do not reconstruct absent original inputs.

Precision on the original loss scale. Table 3 translates the relative target into loss units for the HPM 1 task above, using its frozen forecasts and two queried registers. The six-target loss uses $c_h = 2(b_h - a_h)/6$, so the interval widths are in kWh^2; a 24-target normalization is not used. The targets are stored API-grid increments under the measurement assumptions in Table 4. Exact segment ranges and separately implemented endpoint LPs reproduce the absolute widths.

Model contrast	Initial $W_{r,0}$	Target $W_{r,0}/2$	Terminal W_r	$W_r/W_{r,0}$
Daily–Boosting	0.303143	0.151571	0.001947	0.0064
Daily–MLP	0.194503	0.097252	0.005336	0.0274
Daily–Ridge	0.172075	0.086037	0.000673	0.0039
Boosting–MLP	0.213920	0.106960	0.004617	0.0216
Ridge–Boosting	0.131068	0.065534	0.001766	0.0135
Ridge–MLP	0.107125	0.053563	0.004663	0.0435

Table 3: Original loss scale for the HPM 1 worked example on 5 July 2025, 12:00–18:00 UTC. The first three numeric columns have units kWh^2: they are widths and width targets for identification intervals of six-target MSE differences on stored API-grid increments. The retained endpoint difference is $E = 1.83984375$ kWh. The adaptive policy retrieves the 14:00 register and then the 13:00 register. Normalization uses the initial width of each contrast, not a common numerical loss scale.

What the measurement model certifies. Table 4 separates stored-record guarantees from claims about physical energy. In each mathematical model, the guarantee requires the unknown record to satisfy the declared constraints.

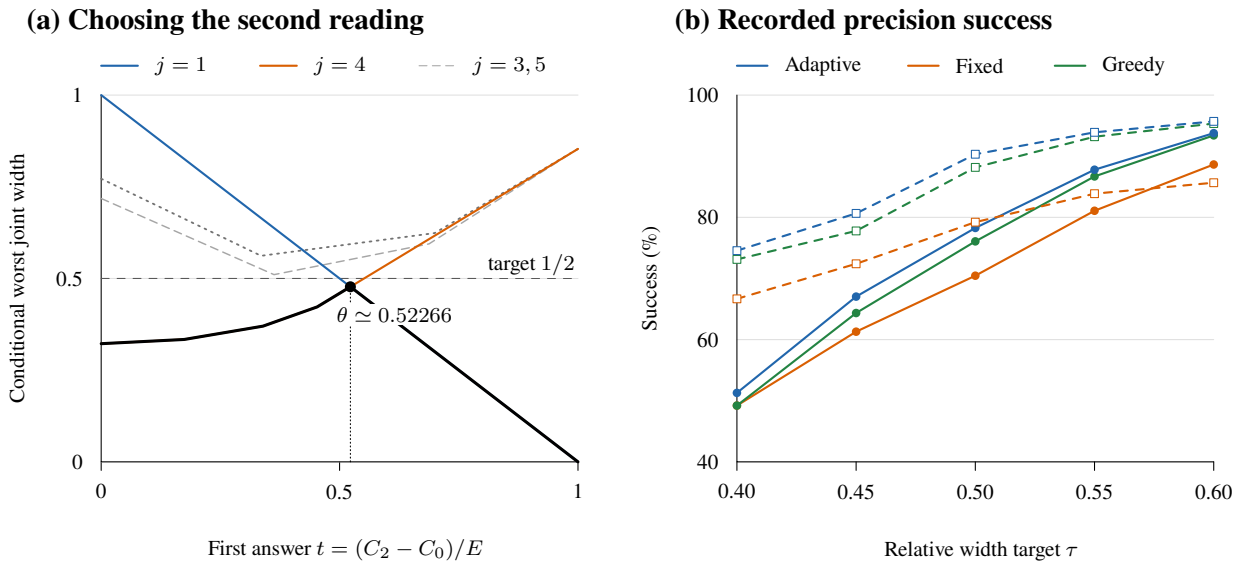


Figure 2: An acquisition rule and its recorded outcomes. (a) Exact conditional envelopes from the saved HPM 1 comparison on 5 July 2025, 12:00–18:00 UTC, after querying boundary 2. Gray curves show the other second-query options; the black lower envelope selects boundary 4 below the switch and boundary 1 above it. Its maximum is about 0.47734, below the half-width target for every possible first answer. (b) The adaptive, fixed-batch and greedy policies on all 819 original and 279 external blocks at five reporting targets. Solid curves represent the original panel and dashed curves the external panel. These exploratory recorded success rates use the existing 10^{-8} comparison tolerance; they are not worst-case precision probabilities.

Table 4: Targets and observation assumptions in the application. Error allowances and hourly caps are sensitivity inputs, not verified device limits.

Model	Target and observations	Scope of the guarantee
Exact stored samples	Targets are increments of stored cumulative samples. Retained endpoints and queried internal samples are exact in this record model.	Certificates concern loss contrasts on the stored grid. They do not establish exact physical energy or historical delivery times.
Bounded internal errors	The latent record includes one fixed internal-error vector with $\lvert e_k \rvert \le \delta E$, where E is the retained block total. Endpoints, and hence the retained total, remain exact. Errors are fixed before policy randomization.	Certificates cover every record and error vector within these bounds. The 1% allowance is illustrative; it is not sampled noise or device calibration.
Physical energy	Actual energy and meter readings would require defensible bounds for internal measurements, endpoints and any physical capacity constraints.	We provide no verified physical calibration. Source meter-class metadata alone does not establish the pointwise error model or exact endpoints.

Unchanged precision under capacity controls. For each shared contrast r, $W_{r,0}$ remains its initial width under the original uncapped exact-total model. The shared cap controls retain $W_r \leq W_{r,0}/2$: they test the same absolute uncertainty budget after adding hourly information. They do not test halving the new capped initial width. The scalar control in Online Resource 1, S6 reports both questions separately. We use the half-width fraction as a declared audit requirement; it is not a calibrated threshold for a downstream decision.

3 Results

The application asks which admissible retrieval method meets a required precision. It separates information available before a query, guarantees over all compatible records, replay on recorded answers, and actual API measurements. All policies in each comparison share the same initial information and measurement model. Scalar readings, network requests and feedback rounds are different resources. The protocol must identify which resource is limited before comparing methods.

3.1 Joint precision with two shared readings

Table 5 separates the planning objectives from recorded precision success. Table 6 identifies where the precision challenger has matching bounds.

Policy	Planning objective	Guarantee and recorded comparison
Exact adaptive	Minimize worst joint width	Optimal among deterministic two-query policies in the declared exact model; replay reports joint precision success.
Fixed batch	Minimize worst joint width	Best deterministic two-query batch for that width objective; not optimized for recorded success.
Greedy adaptive	Myopic first query; optimal continuation	Optimizes the second query after observing the first answer; the first choice need not minimize final worst width.
Randomized mean batch	Minimize worst expected joint width	Batch-optimal for the uncapped exact simplex; replay reports expected width and precision success separately.
Cover + mean fallback	Certify batch precision success	Globally optimal where lower and upper certificates match; elsewhere uses the mean-width policy, with the remaining probability gap reported.

Table 5: Objectives are distinct even when all policies share the same two-reading budget, information and query dictionary. Recorded precision success always means $J \leq 1/2$; it is neither a minimax success value nor proof that a model winner is identified. The noisy adaptive control freezes an exact-model policy and certifies its guarantees anew, without asserting noisy-policy optimality.

Panel	Cases	Closed at 0	Closed at $1/2$	Closed at 1	Open gap	Fallback
Original	819	141	165	162	351	492
External	279	27	57	128	67	94

Table 6: Full-panel status of the batch-precision certificates. "Closed" means equal certified lower and upper bounds on the continuum batch precision optimum. The three closed columns and open-gap column partition each panel. Fallback is a separate, overlapping policy classification: it includes closed-zero and open-gap instances. Thus 468/819 and 212/279 gaps close, but only the 165 and 57 half-success cases provide the nontrivial 1-versus-$1/2$ assurance separation.

Panel / observation model	Blocks	Adaptive	Fixed batch	Greedy
Original, exact	819	641 (78.27%)	577 (70.45%)	623 (76.07%)
External, exact	279	252 (90.32%)	221 (79.21%)	246 (88.17%)
External, 1% allowance	279	242 (86.74%)	215 (77.06%)	235 (84.23%)

Table 7: Recorded-answer joint precision success with two shared readings. We select the fixed batch by its worst-width objective before answers; the last row reoptimizes this comparator for the error model while retaining the coefficient-derived adaptive and greedy policies. The 1% allowance enlarges compatible records around the stored answers; it is not sampled device noise. Endpoints remain exact.

Shared-budget evaluation. Table 5 distinguishes the five acquisition policies. The "closed" and "open gap" labels in Table 6 describe whether the lower and upper precision bounds match. We retrospectively reuse all 819 original and 279 external six-hour blocks, retaining all six positive-width model contrasts in every block. A single pair of cumulative-register queries serves all contrasts. For contrast r, let $W_{r,0}$ denote its original uncapped width and define $J = \max_r W_r/W_{r,0}$. Success means $J \le 1/2$; it is a joint precision requirement, not a guarantee that every model ordering is resolved. Every method receives the same totals, forecasts and two-query allowance. The randomized batch minimizes worst expected J, whereas the deterministic adaptive planner minimizes worst J; it does not optimize over randomized adaptive policies. We compute the threshold and policies without consulting the withheld register responses. The greedy control first minimizes the worst one-query joint width, breaks ties using the best two-query continuation, and then minimizes conditional worst width after the first answer. This one-step refinement baseline uses the same query interface. The original panel contains three BLEM meters over 91 days; the external HeatpumpMonitor panel contains three systems over 31 days. Both use three completed six-hour blocks per day and the same four frozen model types: daily persistence, ridge regression, gradient-boosted trees and a multilayer perceptron. The declared intraday audit supplies block endpoints and day-start forecasts before optional retrieval; source and timing details appear in Sections 3.3 and 3.2.

The adaptive success gains over the best fixed batch for the worst-width objective are 7.81, 11.11 and 9.68 percentage points, respectively. Its gains over greedy are smaller: 2.20, 2.15 and 2.51 points. For the exact external panel, the mean realized J is 0.2823 for adaptive, 0.3381 for the fixed batch and 0.2893 for greedy. At the 1% allowance these are 0.3028, 0.3481 and 0.3099. The randomized mean-width batch has a different objective; its recorded precision probability is not

an optimal-batch precision baseline. We retain all block-level outcomes, including unfavorable comparisons.

The cover + mean fallback policy uses a guaranteeing fixed batch when available, then a certified equal mixture of two batches, and otherwise a mean-width fallback. Its recorded success is 54.98% over the original panel and 72.55% over the external panel; these full-panel rates are not those of a universally optimal precision policy. Matching certificates prove a batch precision optimum of $1/2$ on 165 original and 57 external instances. On these instances, adaptive succeeds on every stored record, while the optimal batch mixtures succeed with expected frequencies 108/165 (65.45%) and 37/57 (64.91%). Coefficient certificates define these subsets before record replay. The complete fixed-batch and greedy comparisons in Table 7 remain the broader controls.

Capacity control. For an illustrative cap $y_h \leq E/2$, we reoptimize the noiseless planner using concave piecewise-linear segment widths. We solve each conditional problem by checking all line intersections within each capacity interval. We do not infer a capped randomized-batch optimum from simplex vertices. Of the 819 original and 279 external records, respectively 780 and 200 satisfy this cap. Among those compatible records, adaptive-only half-width guarantees remain in 56 and 4 blocks; recorded adaptive versus fixed-batch successes are 655 versus 627 and 179 versus 177. Under the stronger cap $E/3$, only 613 original and 198 external records are compatible. No adaptive-only half-width guarantee remains, and the two planners have equal success counts of 613 and 192. We flag incompatible records and make no cap-based guarantee for them. Thus supplied capacity information can remove the need for feedback. Combining the $E/2$ cap with the 1% stored-reading allowance, the capacity-aware frozen adaptive policy succeeds on 169/200 compatible records, versus 166/200 for the best fixed batch under that model. Mean joint widths are 0.3479 and 0.3535. This is a small observed gain, not evidence of uniform operational superiority; the frozen uncapped greedy control succeeds on 165/200 and is not presented as an optimized cap-aware comparator.

Paired uncertainty and source variation. The external exact-panel advantage over the worst-width fixed batch is 11.11 percentage points (descriptive 95% interval 8.60–13.62), compared with 2.15 points over greedy (1.08–3.23). These percentile intervals use 2,000 seven-day circular calendar-block resamples, synchronized across the three systems and intraday blocks, with seed 20261013. The combined cap/error gain over the fixed batch is 1.50 points (-0.49–3.16). Paired 2×2 counts, individual-system effects and one-day-cluster sensitivity appear in Online Resource 1, S8. The intervals describe these panels; three systems cannot establish population-level generalization, and dependence beyond the resampled blocks remains unmodeled. Table 8 shows the system-level paired counts. The external gain over fixed batching is heterogeneous: system 2 supplies 25 of the 31 net additional successes, while system 7 has one fewer success than greedy adaptation. Omitting each system in turn leaves positive exact-model panel effects: 5.13–10.07 and 3.23–16.13 percentage points over fixed batching for the original and external panels, respectively; the corresponding ranges over greedy are 1.65–3.11 and 1.08–3.76 points. A fourteen-day synchronized circular-block sensitivity also retains positive exact-model intervals (Online Resource 1, S8). These checks reuse the same observations; they are not additional independent replications. The primary seven-day cap-plus-error interval remains $[-0.49, 3.16]$ points. Its fourteen-day sensitivity is $[0.48, 2.63]$, and its

System	n	A / F / G	A-only/F-only	Δ_F (pp)	A-only/G-only	Δ_G (pp)
O1	273	212/193/211	27/8	6.96	3/2	0.37
O2	273	196/187/188	30/21	3.30	9/1	2.93
O3	273	233/197/224	38/2	13.19	12/3	3.30
HPM 1	93	76/75/73	3/2	1.08	3/0	3.23
HPM 2	93	91/66/87	25/0	26.88	4/0	4.30
HPM 7	93	85/80/86	5/0	5.38	0/1	-1.08

Table 8: Exact-reading paired joint-precision outcomes by system. A is the exact deterministic adaptive planner; F is the best deterministic batch for worst width; G is greedy adaptation. The two discordant counts identify blocks where only A or only its comparator succeeds. O1–O3 are BLEM consumers 1–3; HPM labels retain provider system identifiers. Blocks are repeated observations within systems.

leave-one-system-out point effects range from 0.81 to 2.63 points. This small three-success gain remains sensitive to the temporal specification; the favorable secondary interval does not replace the primary assessment.

Checking batch-precision guarantees. For each selected batch pair, 36 rational multiplier checks prove that the two batches cannot both fail on any feasible record. Feasible adverse records give matching upper bounds, establishing the full-continuum $1/2$ optima. Rational checks also verify the adaptive half-width guarantees. Elsewhere, we report the gap between the challenger's lower and upper bounds; the mean-width fallback has only the trivial precision lower bound zero. Online Resource 1, S8 gives all multipliers, adverse completions and coefficient-based tie rules. These certificates assume exact readings; the next calculation treats errors separately.

When feedback changes the available guarantee. Exact rational checks partition the original 819 blocks into 162 where a fixed batch guarantees half-width, 168 where adaptive querying does so but no deterministic batch does, and 489 where no deterministic two-query policy does. The corresponding external counts are 128, 57 and 94. These are guarantees over the declared uncapped, noiseless simplex, not frequencies of success on the recorded trace.
Among the 165 original and 57 external blocks with matched randomized-batch optimum $1/2$, the greedy first query fails the half-width requirement on 25 and 13 blocks, respectively, even with its optimal conditional second query. Greedy nevertheless succeeds on the recorded trace in 20 of those 25 and 11 of those 13 blocks. The diagnostic identifies a specific benefit of planning the first query: recorded success can conceal an inadequate pre-query guarantee. Across the full panels, this first-query limitation occurs on 28 and 16 adaptive-guaranteed blocks. The supplied interval covers and uncovered-answer witnesses use rational arithmetic on the stored contrast coefficients.

A stronger fallback check. We retain the certified deterministic or equal two-batch cover and replace its mean-width fallback by the fixed worst-width batch on the same 492 original and 94 external fallback blocks. Using the strict threshold $J \leq 1/2$, this coefficient-only change raises recorded success from 54.98% and 72.55% to 64.71% and 74.19%.

Pure fixed batching still performs better (70.45% and 79.21%); the entire recorded-score deficit arises on the matched-half subsets, where fixed batching succeeds on 155/165 and 51/57 records, versus expected cover successes of 108/165 and 37/57. On each of those instances, however, every fixed batch has worst-case precision success zero, whereas the cover guarantees $1/2$. The matched-half guarantees and open continuum gaps also apply to this variant. This retrospective diagnostic reuses both panels; the supplement gives both threshold conventions, block-level paired outcomes, system summaries and descriptive seven-day block-bootstrap intervals.

How much remains unresolved in the batch comparison. The batch-precision cover has matching continuum bounds on 468 of 819 original blocks and 212 of 279 external blocks. On the remaining 351 and 67 blocks, the median upper-minus-lower probability gap is respectively $1/3$ and $1/2$. The interquartile ranges are $[1/4, 1/2]$ and $[1/3, 1/2]$; both panels range from $1/5$ to $1/2$. Averaged over all blocks, including closed gaps, the gaps are 0.1583 and 0.0983. These unresolved gaps are substantial. We therefore use the 165 original and 57 external matched-half certificates as the evidence of exact precision superiority; the full-panel cover replay is a comparison with a declared policy, not with a globally solved precision optimum.

Sensitivity to the reporting target. Figure 2(b) replays the same policies and all 1,098 blocks at relative width targets $0.40, 0.45, 0.50, 0.55, 0.60$. This post-hoc sensitivity analysis is exploratory; we did not select new queries, refit models, or tune policy mixtures for the new targets. Across these targets, the adaptive policy's recorded advantage over fixed batching ranges from 2.08 to 7.81 percentage points on the original panel and from 7.89 to 11.11 points on the external panel. The advantages over greedy range from 0.37 to 2.69 and from 0.36 to 2.87 points. Some descriptive seven-day-block intervals for the greedy comparisons include zero. Thus the fixed-batch difference persists around the declared half-width target, while the incremental value over greedy remains smaller.

Shared-query sensitivity to stored-reading errors. Across all 279 external blocks, the frozen shared policy guarantees that all six normalized comparison widths are at most $1/2$ on 165 blocks under fixed internal-reading errors bounded by 1% of block energy; the best deterministic batch does so on 120. On the 45 additional blocks, feasible fixed-record certificates prove that every randomized batch has worst-case precision success below one. The certified batch success upper bound has minimum and median $1/2$, interquartile range $[1/2, 2/3]$, and maximum $4/5$; 26 bounds are at most $1/2$ and 37 are at most 0.7. These are upper bounds on the unrestricted batch optimum, not claims that each bound is attained. The 45 blocks span all three external systems (11, 11, and 23 blocks).

Capacity information reduces the benefit. With a per-hour cap of half the block energy as well as the same 1% error allowance, a policy frozen from the capacity-aware noiseless planner certifies 213 of the 279 coefficient instances, versus 204 for the best deterministic batch; nine have a certified randomized-batch limitation, with upper bounds between $2/3$ and $7/8$. Only 200 recorded profiles satisfy this illustrative cap. Within that compatible subset the counts are 153 versus 147, with six separations. The original uncapped policy certifies only 191 of the 279 capped instances, which motivates using declared capacity information when planning. Both

policies are fixed before the fine readings are opened; neither is claimed optimal for bounded-error acquisition.
The comparison uses two shared readings, uncapped noiseless initial widths for normalization, exact block endpoints, and the same permitted queries for every method. A separately implemented rational checker verifies the saved contrast coefficients, branch coverage, every reported precision status, and the complete fixed-error adverse records. Supplementary methods (Online Resource 1, S8) give the certificate construction.

Solver and information-flow checks. All policies reuse the same two counter answers across the six contrasts. Plans are saved before recorded answers are loaded; this verifies the new run's dependency order, not prospective independence of the reused data. A separate implementation reconstructs every plan and summary on all 1,098 blocks. On 18 fixed representative blocks it solves 19,440 direct endpoint LPs and 6,480 conditional LPs. Maximum discrepancies are 2.22×10^{-16} and 1.11×10^{-16}; the numerical check tolerance is 10^{-8}. The largest batch primal–dual gap is 2.03×10^{-10}. The numerical adaptive worst-width value is below the optimal randomized batch mean value on 462 original and 162 external blocks. This is a subset result, not uniform dominance; the full-panel mean differences are only 0.00177 and 0.00393 of initial width. An outcome-informed diagnostic optimizes recorded outcomes over the mean-optimal batch face, allowing 10^{-9} objective slack. Even its best precision success is 54.285512% original and 69.898131% external, essentially the declared mixtures' 54.285508% and 69.898130%. Thus arbitrary LP tie selection does not explain this comparison. These diagnostic oracle choices are not deployable acquisition policies, nor optimal batches for worst-case precision success. S8 supplies all ledgers, per-meter results, exact bounded-error certificates and reproduction commands.

Source-to-table verification. A separate execution regenerated the original counter inputs from three version-pinned source files. Restoring the 6,552 omitted target and persistence rows recovered the exact byte hash of the 89,496-row frozen forecast archive. The external check verified all cached provider decimals against the hourly export. Source-to-forecast joins checked all 6,588 contrasts across the 819 original and 279 external blocks; maximum coefficient discrepancy was below 1.12×10^{-16}, and retained totals agreed exactly. Fresh policy and certificate generation reproduced the 165 and 57 matched half-optimal cases. Exact checks covered 7,992 covering multipliers and 463 adaptive branch intervals. An independent replay recomputed all 10,980 recorded batch widths and 65,880 paired model-ordering checks. All 1,098 complete-panel outcome rows agreed with the reported tables, with zero numerical difference. Supplemental files retain commands, hashes and execution reports; raw-dependent original records are regenerated locally. This verifies the frozen-predictor computations, without refitting models or claiming external human replication.

3.2 Source protocols and the separate-pair controls

The original source contains three cumulative consumption counters (Kostmann and Härdle, 2019), with 91 eligible days per meter. The four frozen forecasters are daily persistence, ridge regression, boosted trees and an MLP. We audit six-hour blocks ending at 06:00, 12:00 and 18:00

UTC using block endpoints and day-start forecasts. Next-day persistence copies are not yet available under this declared schedule; production arrival timestamps are not measured. The interface permits five internal hourly counters. A two-reading budget per separate-pair audit gives 4,914 tasks; a two-reading budget shared by all pairs gives the 819-block study above. These are different resource requirements. Full source and fitting specifications are in Online Resource 1, S3; S6 retains the complete separate-pair controls.

In the separate-pair experiment, a fixed batch guarantees half width on 4,371 tasks, adaptation is required for the deterministic guarantee on 290, and 253 have no such deterministic two-query guarantee. All 290 have exact randomized-batch precision optimum $1/2$. Recorded adaptive success is 98.64%, versus 98.23% for the best fixed worst-width batch and 97.96% for greedy; the adaptive–fixed difference is only 0.407 percentage points. The prespecified 10% feedback-prevalence target was not met (5.90%). Adding an illustrative hourly cap of $E/2$ leaves only eight of the 290 cases requiring feedback at the uncapped half-width threshold; an $E/3$ cap leaves none. Recorded profiles violate those caps in eight and 57 cases, respectively. S6 retains both the original-threshold and new-constrained-width calculations, all compatibility flags and witnesses.

3.3 External source and bounded-error controls

The HeatpumpMonitor source protocol (OpenEnergyMonitor, 2026a,b) selected the first three ascending system IDs with public MID status, explicit Class-1 electrical-meter metadata and a cumulative feed spanning January–July 2025. Systems 1, 2 and 7 qualified. The source rule preceded hourly retrieval. Training gates, set after inspecting training data but before July retrieval, required 95% valid hourly intervals and 90 complete days. All three passed without replacement. January–June supplied training; July supplied 31 test days per system. Each of the four model specifications was fitted once, without retuning. The source, exclusion and fitting chronology is retained in S6. The later shared-query analysis reuses these sources and is retrospective.

Targets are increments of stored API-grid cumulative samples. The interface may label a selected native sample with a grid timestamp (OpenEnergyMonitor, 2026c); native acquisition and historical delivery times are unavailable. The stored-record interpretation follows Table 4. Class-1 metadata is an eligibility condition, not a pointwise error bound.

All 1,674 separate-pair tasks (279 shared blocks) remain eligible. A fixed batch guarantees half width on 1,614, feedback supplies the deterministic guarantee on 29, and 31 have no such two-query guarantee. The 29 span every system and model pair and have exact batch precision optimum $1/2$. The adaptive–fixed recorded gain over all tasks is only 0.119 percentage points; greedy succeeds on all 29 selected cases. S6 retains the complete comparisons and exact checks of 16,740 terminal intervals.

Error allowances and required budgets. With exact endpoints and fixed internal-readout errors bounded by $0.001E$ or $0.01E$, respectively 27 and 22 of those 29 cases retain the two-query adaptive guarantee. Both levels cover every system and model pair. Exact lower witnesses rule out a deterministic two-query guarantee for the remaining cases; no randomized-adaptive impossibility is claimed. A separate certified rule uses two adaptive readings for 22 cases and a fixed three-reading batch for seven at $0.01E$, guaranteeing precision in all 29. Its 65 reserved

readings compare with 87 for fixed batches, but require 51 information rounds rather than 29. This is a post-analysis per-pair hard-budget result, not expected stopping cost or a shared-query result. S6 provides the complete latent-error model, rational branch and lower witnesses, and three-reading certificates. The next section tests the provider's actual unrestricted interface.

3.4 Retrieval interface and measured operational consequences

The public source permits both single-time and full-range queries (OpenEnergyMonitor, 2026b,c). We therefore tested the latter as an operational comparator instead of treating a two-reading budget as a measured tariff. The first eligible daily-persistence–MLP task from each external system was fixed before retrieving optional answers. Two repetitions of five policies produced 30 trials and 60 actual HTTP requests, including six common endpoint reads. The protocol limited request starts to one per second, with no retries or cache-busting. All trials succeeded; timings include the recorded persistent-connection behavior and uncontrolled public cache/server state.

Table 9: Actual public-API measurements on three fixed tasks, two repetitions per policy. Incremental I/O excludes the separately logged client throttle wait. Bytes are response-body bytes, not wire traffic. The full-range interface returns seven hourly samples in one request, including retained endpoints; it is an operational alternative outside the two-optional-sample benchmark budget.

Policy	Requests	Mean I/O (s)	Mean body bytes
Exact adaptive	2	0.497	104.3
Best deterministic worst-width batch	2	0.539	104.7
Mean-width-optimal batch	2	0.582	104.3
Greedy adaptive	2	0.551	104.3
Full hourly range	1	0.331	226.7

All five policies attained the precision target on these three traces. Bulk retrieval used fewer requests and less observed I/O time, while also reducing terminal width to zero. Selective retrieval reduced body size by about 122 bytes; this small difference excludes HTTP/TLS overhead and does not establish an operational saving. A range returning only the five internal samples could reduce the bulk response further. Including the imposed client throttle, mean completion times were 1.984 seconds for exact adaptation and 1.040 seconds for bulk retrieval; these are not intrinsic provider latencies. The small workload does not support a population speedup or monetary-cost claim.

An offline checker matches all 60 received payload hashes to separately cached source responses, checks all 12 adaptive second-query decisions and all 30 interval containments (maximum discrepancy 1.74×10^{-18}). Each result can be verified without repeating provider traffic. For this observed interface, select bulk retrieval when its unrestricted response is admissible.

Selecting a feasible acquisition protocol. The reference implementation chooses among supplied certified protocols using the query interface, scalar-reading and information-round limits, and the requested priority. On the same 29 external pairwise cases at 1% internal error, it selects

bulk in all 29 when range retrieval is admissible; under a three-reading restricted interface it uses two adaptive readings in 22 cases and fixed-three in seven. A one-round restriction instead selects fixed-three in all 29. Restricting both interfaces to two readings leaves seven cases requiring escalation. The selector does not relax precision or treat a missing catalogue certificate as a general impossibility result.

The operational wrapper binds each certificate to its model pair, coefficients, meter, block, retained endpoints and source hashes. It rejects readout envelopes for a different task, timestamp, feed or unit. Online Resource 1, S8 retains all 116 protocol-selection decisions, callback tests, exclusions and separate task-binding checks. These protect against accidental mismatch; they do not authenticate a provider or calibrate a meter.

Reading charges and feedback delays favor different protocols. If a reserved reading costs one unit and an information round costs $r \geq 0$, the 29-case fixed-three plan costs $87 + 29r$, while adaptive-where-certified with fixed fallback costs $65 + 51r$. The latter is preferable for $r < 1$ and fixed-three for $r > 1$. This exact catalogue frontier is a declared-cost sensitivity, not a measured saving. S8 retains the full grid and all 306 independently checked candidate evaluations. Actual public-API retrieval favors bulk: the measured three-task mean I/O times are 0.331 s for bulk and 0.497 s for adaptive retrieval. A shared-query objective does not change that interface advantage.

3.5 Computability and proof checks

The proofs establish the universal classification. Numerical checks address the implementation separately: 3,202 endpoint LPs and 28 finite games agree with the analytic constructions, with largest width discrepancy 3.34×10^{-16} against a 2×10^{-8} scale-aware tolerance. Six explicit polytopes check overlapping constraints, signed contrasts, boxes and coupled capacities. All 37 enumerated batches and 24 possible first queries agree with the corresponding LP guarantees. These are small formulation checks, not a general scalability claim.

For coordinate queries under disjoint exact totals and unit costs, a supporting compact mean-width solver avoids enumerating batches. Its retained-endpoint flow representation and feasible dual completion and adaptive lower-bound proof are given in Online Resource 1, S5; S7 supplies the measured comparisons. General weighted-oracle and dynamic-program LP constructions are established (Kawase and Sumita, 2024; Martin et al., 1990). The specialized representation and its hard-budget policy decomposition are the implementation contribution.

All 28 archived compact settings solved. Twelve prescribed enumerated comparisons agreed within 1.27×10^{-15} after normalization. Compact was faster in eight and used less whole-process peak memory in six; enumeration was faster in the four smallest one-block settings. The largest tested LP had 38,820 variables and ran in 9.933–10.012 seconds for 48 target coordinates. These are single-run measurements on a shared Intel Core i7-8565U host with one numerical thread. The complete timing, memory, residual, tie-profile and skip ledgers are in S1/S7. No large-scale optimizer for arbitrary convex sets or precision risk is implied.

3.6 Historical controls: guarantees and observed accuracy

The completed-archive control (Trindade, 2015; Hebrail and Berard, 2006) incorporates every permitted forecast copy before evaluation. Under identical information and zero new readings, paired bounds certify 7,836/9,720 Portugal comparisons (80.62%), versus 2,513 (25.85%) from

separate-loss bounds. However, a donor imputer makes no observed whole-period errors at full coverage; the added value is an all-completions certificate, not demonstrated prediction accuracy. In the daily archive analysis, free reconstruction already resolves 95.33% of masked targets, exact acquisition adds only 0.042 percentage points over greedy, and robust randomization worsens realized width despite improving its objective. Outcome-independent optimal-face refinements retain that tradeoff (Iancu and Trichakis, 2014). The coordinate-acquisition control reports a 0.9738064% mean acquisition saving. S2–S4 give its denominator, these adverse controls, all 324 grid cells, four-model gates, bootstrap details and full protocols. They explain why the present paper evaluates the assurance requirement and retrieval interface explicitly rather than claiming general acquisition savings.

4 Discussion

4.1 What the method changes in an audit

The auditor first states the contrast, required precision, available information and retrieval options. A sufficient initial interval needs no query. Otherwise, a bulk response or fixed batch can avoid feedback; adaptive retrieval is useful when it meets the required guarantee with fewer permitted readings.

What the application establishes. With two readings shared across six comparisons, feedback guarantees the required precision on 165 original and 57 external blocks where every randomized fixed batch has worst-case success at most one half. Matching batch constructions attain that bound. This is a change in the assurance available at the same budget. The worked example also reports the six interval widths in original loss units (Table 3). These precision certificates complement the constructed mean-risk theorem; they do not measure its $1/2$-versus-$2/3$ gap on the data. Across the complete panels, recorded gains over the best fixed worst-width batch are 7.81 and 11.11 percentage points, with smaller gains of 2.20 and 2.15 points over greedy adaptation. We do not infer improved forecasting accuracy or operating savings from these precision results. The separate-pair replay has smaller recorded gains of 0.407 and 0.119 percentage points. Both analyses use three systems per source; repeated days and model pairs do not establish population prevalence.
The capacity controls show that feedback's larger gains need not survive additional hourly information, while internal-reading errors can increase the sufficient budget. These are alternative information models, not deployment-calibrated bounds (Table 4). The half-width target limits the remaining range of each loss contrast; we have not validated it as a threshold for financial action or model selection. A secondary shared-query check certifies a best model on 123 original and 207 external blocks, versus 122 and 201 for fixed batches and 118 and 208 for greedy adaptation. On the external panel, greedy certifies one more best model despite its lower joint-precision success. The width objective therefore does not uniformly improve model choice. Requests until a decision are another objective; neither theorem optimizes it.

Choosing an acquisition interface. Bulk uses one request and is faster on the measured three-task API workload, so the implementation selects it when admissible. Under the separate

restricted-reading contract, it uses 22 two-reading and seven three-reading prescriptions. Their 65 reserved readings compare with 87 for fixed batches, but they require 51 information rounds rather than 29. The declared cost frontier explains when each prescription is preferable; it is not measured financial or service-wide benefit. Capacity bounds can likewise make fixed retrieval sufficient. The practical output is a checked choice under a stated contract, including cases where selective adaptation is unnecessary.

Breadth of the theory and limits of the implementations. The scalar classification permits compact convex sets, affine targets, a finite linear-query dictionary and answer-independent legal query sets. It includes signed targets, overlapping constraints and unequal fixed costs. Fixed bounded errors can enter the hidden record (Online Resource 1, S6). The finite dictionary matters: unrestricted linear measurements would permit querying the scalar target directly. The algorithms are more specialized: general polyhedra with an enumerated batch family or two-query planning; cumulative sums with a known nonnegative total; and compact coordinate recovery with disjoint exact totals and unit costs. They do not provide an efficient arbitrary-budget precision optimizer for every convex set.

5 Conclusions

The risk objective determines whether feedback can improve the worst-case guarantee for a model-comparison interval. Concavity gives a complete no-improvement classification; nonconcave penalties admit sharp query-based separations. Simultaneous comparisons have a different boundary: the largest normalized width can benefit from feedback even under a mean-risk objective. An exact two-query rule and checkable certificates turn that distinction into an acquisition decision. The reference implementation chooses among supplied bulk, fixed and adaptive certificates using declared access and resource limits. New comparisons require new certificates; a catalogue exclusion alone is not an impossibility proof. The public-data results show both when feedback changes assurance and when bulk or fixed retrieval is preferable. These choices remain conditional on the declared uncertainty set and access rules.

Supplementary Materials

Online Resource 1 is supplied as Supplemental Data S1 and S2. Extract both ZIP files into the same directory. Together they contain the editable manuscript, executable methods, frozen forecasts, source identifiers, case ledgers, exact witnesses and numerical checks. Start with `README.txt`; `CURRENT_RESULTS_MAP.txt` maps each current table and verification claim to its command, inputs and outputs. The default verification replays saved forecasts and does not train new models. Separate commands cover raw-input regeneration and fresh fitting. BLEM-derived traces are reconstructed locally; attributed HeatpumpMonitor records are bundled. The indexed extended methods retain full proofs, solver scaling, historical controls, forecasting protocols and uncertainty checks. Protocol-selection files include all 116 interface decisions and task-bound readout checks. The separate checker compares saved contrasts with frozen geometry, verifies

forecast hashes, and reconstructs retained endpoints from cached API records. Historical reports relevant to this revision are retained; the larger development archive is preserved separately.

Additional Information and Declarations

Data and Code Availability

The numerical code and detailed outputs are supplied in Online Resource 1. The public source identifiers are given in the references. The package separates bundled Portugal/household replay from BLEM raw-source regeneration. BLEM counter traces are regenerated from the pinned public QuantLet sources; the commands reject inputs with mismatched hashes. These raw traces are not redistributed because redistribution permission has not been established. Constructed polytopes include their complete inequalities and objectives. Dual outputs retain raw multipliers, any numerical mass correction, the completed record, and pricing and enumerated lower bounds. LP and endpoint cross-checks use floating-point arithmetic with explicit tolerances; the counter covering and adverse-profile certificates are additionally checked with exact rational arithmetic on their serialized inputs. Shared-query checks include all joint outcomes, optimal-batch tie diagnostics, bounded-error dual certificates and adverse completions. Reviewers receive the current executable materials as Supplemental Data S1 and S2.

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Competing Interests

The author declares no competing interests.

Author Contributions

Jishan Kapoor defined the research objectives and evaluation requirements, directed and supervised the analysis and revisions, interpreted the findings, and approved the manuscript. The computational and drafting assistance used under this supervision is described below.

Research Transparency

This work uses public data and constructed mathematical instances, with no newly recruited participants. New measurements concern public API response times and payloads, with their small-workload scope reported. Retrospective masks and archive-access assumptions are identified explicitly. Comparisons use the same information and the same stated hard budgets. The new proofs and numerical checks do not change the historical forecast results.

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